

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

11th September 2021

Subject SP8 – General Insurance: Pricing

Time allowed: 3 Hours 30 Minutes (14.30 – 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. *Please read the instructions to examinees sent along with hall ticket carefully and follow without exception.***
- 2. *The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.***
- 3. *Mark allocations are shown in brackets.***

- Q. 1)** i) State the different reinsurance basis for writing reinsurance policies. (1)
- ii) List and differentiate between run-off reinsurance solutions available to an Insurer. (4)
[5]
- Q. 2)** XYZ is a large General Insurance company writing commercial property insurance. The underwriter currently charges premium based on documented base rate and adjust those depending on risk factors. The Company has witnessed increased loss ratio in recent years.
- i) List the rating factors which can be used by underwriters to adjust the base rate. (5)
- ii) Outline the Actuarial investigations that need to be performed to understand the reasons for rising loss ratio and suggest pricing approach to arrive revised base rates. (12)
[17]
- Q. 3)** i) Actuarial team is using a GLM model to determine premium for motor insurance product. The team is divided on whether to use "Experience" and "Age of the driver" as a rating factor. Briefly explain how Chi squared statistic, F- statistic and AIC can be used to ascertain the relevance of rating factors. (6)
- ii) Differentiate between Lift Curve and Gains curve for validating the model used to determine the premium for the product stated in (i) above. (4)
[10]
- Q. 4)** The Insurance Regulator has finalized new guidelines in respect of Standard Fire product for Dwellings and has mandated all Insurance companies to offer the same. You are the pricing actuary of XYZ medium sized insurer operating in the Country with fire as one of the major Line of business.
- i) Describe how FLEXA and STFI risks can be priced for such product. (5)
- ii) After launching of product, you have got feedback from the marketing team that premium for the product is high compared to the competitors. Your CEO is asking you to provide a report on why the premiums would be different among insurers though the coverage is the same. State the reasons to be mentioned in the report. (4)
- iii) Marketing team of XYZ insurer is worried that the take-off of the product may be lower in view of high premium. Suggest ideas for increasing the sales of the product despite high premium. (2)
- iv) XYZ insurer is also planning to launch add-on covers to be offered for the same product i.e. EMI Protect and Temporary resettlement for occupants of the house. Explain the considerations while pricing the add-ons. (3)
[14]
- Q. 5)** Country has witnessed Covid-19 pandemic and Government has imposed partial/full nationwide lockdowns to curb infection.
- i) List possible impact on health insurance claims on account due to pandemic and Government actions. (2)

A leading private Health Insurance Company underwrites both Retail and Group Indemnity Health portfolio. The Company uses experience rating to determine premiums for a particular group whereas retail portfolio pricing is basis a rate chart which varies as per age, sum insured and type of plan chosen by individual. The CFO is worried about surge in claims costs from pandemic and have suggested to increase premiums by 25% for both retail and group.

ii) Discuss the investigations a Pricing Actuary need to perform to evaluate the CFO's proposal. (10)

iii) Discuss other additional factors which could influence premium charged. (5)
[17]

Q. 6) **i)** Explain how to price an Aggregate Excess of Loss treaty for property line of business. (8)

ii) Explain how an indexation clause can be considered in (i) above. (2)
[10]

Q. 7) ABC is newly incorporated multi-line general insurer and aspires to develop a robust and industry leading data system for monitoring renewal rate and premium charged for risk underwritten. The company offers short term, annual and multi-year policies on one time, annual and EMI option for premium payments.

i) List the data required to monitor the renewal rate and average premiums by giving examples. (4)

ii) Identify the key features required to make an industry leading data system. (10)
[14]

Q. 8) Discuss why burning cost method may not be suitable to price large commercial risk and how catastrophe model could be useful in such cases. [8]

Q. 9) You are assisting the Chief Actuary in pricing a pet insurance product.

i) Describe the expense heads that may be considered in pricing this product. (2)

ii) List the steps involved in estimating expense loading for arriving at office premium. (3)
[5]
